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Random Number generation

Inverse Transform Sampling, Exponential Distribution,
Q-Q Plots, Cauchy Distribution

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Chapter 1

Inverse Transform Sampling Method

1.1 Introduction

The **Inverse Transform Sampling** is a fundamental technique for generating random numbers from any continuous probability distribution. Given a source of uniform random numbers, this method allows us to generate samples from an arbitrary distribution with a known cumulative distribution function (CDF).

1.2 Mathematical Foundation

The theoretical foundation rests on this fundamental theorem:

Theorem: Let X be a continuous random variable with CDF $F(x) = P(X \leq x)$. Then the random variable $U = F(X)$ follows a $\text{Uniform}(0, 1)$ distribution.

Conversely, if $U \sim \text{Uniform}(0, 1)$, then $X = F^{-1}(U)$ has the probability distribution corresponding to F .

1.3 Proof Sketch

For $0 \leq u \leq 1$:

$$P(U \leq u) = P(F(X) \leq u) = P(X \leq F^{-1}(u)) = F(F^{-1}(u)) = u$$

This is exactly the CDF of a $\text{Uniform}(0, 1)$ distribution.

1.4 Algorithm

The algorithm is straightforward:

1. Generate $U \sim \text{Uniform}(0, 1)$
2. Compute $X = F^{-1}(U)$
3. The sample X follows the desired distribution

4. Repeat n times to generate n samples

In equation form:

$$X = F^{-1}(U), \quad \text{where } U \sim \text{Uniform}(0, 1)$$

1.5 Why It Works

The validity follows directly from the probability integral transform:

$$P(X \leq x) = P(F^{-1}(U) \leq x) \quad (1.1)$$

$$= P(U \leq F(x)) \quad (1.2)$$

$$= F(x) \quad (1.3)$$

This confirms that the generated samples have exactly the desired CDF.

1.6 Pareto Distribution Example

The Pareto distribution is commonly used to model phenomena with heavy tails, such as wealth distribution, firm sizes, and city populations.

For the Pareto distribution with minimum value $x_m > 0$ and shape parameter $\alpha > 0$:

$$\text{PDF: } f(x) = \frac{\alpha x_m^\alpha}{x^{\alpha+1}}, \quad x \geq x_m \quad (1.4)$$

$$\text{CDF: } F(x) = 1 - \left(\frac{x_m}{x}\right)^\alpha \quad (1.5)$$

$$\text{Mean: } E[X] = \frac{\alpha x_m}{\alpha - 1}, \quad \alpha > 1 \quad (1.6)$$

$$\text{Inverse CDF: } F^{-1}(u) = \frac{x_m}{(1 - u)^{1/\alpha}} \quad (1.7)$$

Derivation of inverse CDF:

Starting with:

$$u = F(x) = 1 - \left(\frac{x_m}{x}\right)^\alpha$$

Solving for x :

$$\left(\frac{x_m}{x}\right)^\alpha = 1 - u \quad (1.8)$$

$$\frac{x_m}{x} = (1 - u)^{1/\alpha} \quad (1.9)$$

$$x = \frac{x_m}{(1 - u)^{1/\alpha}} \quad (1.10)$$

Generation algorithm:

Since $1 - U$ is also uniformly distributed on $(0, 1)$ when $U \sim \text{Uniform}(0, 1)$, we can simplify:

$$X = \frac{x_m}{U^{1/\alpha}}$$

where $U \sim \text{Uniform}(0, 1)$.

1.7 Advantages and Disadvantages

Advantages:

- Works for ANY distribution with known/computable inverse CDF
- Efficient: no rejection sampling, all uniforms are used
- Direct transformation: $\mathcal{O}(1)$ time per sample
- Can generate correlated samples by transforming same uniforms
- Simple and elegant

Disadvantages:

- Must know or compute $F^{-1}(u)$ analytically or numerically
- For some distributions, computing F^{-1} is expensive
- Can be numerically unstable near tails ($u \rightarrow 0$ or $u \rightarrow 1$)
- Not applicable when inverse CDF cannot be computed

Chapter 2

Comparison with LLM Explanations

The LLM model that is used is the simple version of ChatGPT. Also, one of the requirements was to write the explanation in LaTeX.

Inverse Transform Method for Random Number Generation

The inverse transform method is a technique used to generate random numbers distributed according to a desired probability distribution, starting from a uniform random variable.

Step 1: Start with a Uniform Random Variable

Let U be a random variable uniformly distributed on the interval $(0, 1)$:

$$U \sim \mathcal{U}(0, 1)$$

Step 2: Define the Target Distribution

Let X be a continuous random variable with probability density function (pdf) $f_X(x)$ and cumulative distribution function (CDF):

$$F_X(x) = \int_{-\infty}^x f_X(t) dt$$

We assume that $F_X(x)$ is strictly increasing and therefore invertible.

Step 3: Apply the Inverse CDF

Define a new variable X as:

$$X = F_X^{-1}(U)$$

Step 4: Derivation of the Result

We show that X has the desired distribution.

Compute the CDF of X :

$$\begin{aligned}P(X \leq x) &= P(F_X^{-1}(U) \leq x) \\&= P(U \leq F_X(x)) \\&= F_X(x)\end{aligned}$$

Since $U \sim \mathcal{U}(0, 1)$, we used:

$$P(U \leq y) = y \quad \text{for } y \in [0, 1]$$

Thus:

$$P(X \leq x) = F_X(x)$$

which shows that X indeed follows the desired distribution.

2.1 Summary

The inverse transform method consists of:

1. Generate $U \sim \mathcal{U}(0, 1)$
2. Compute $X = F_X^{-1}(U)$
3. Then X follows the distribution with pdf $f_X(x)$

Remarks

- The method requires that the CDF $F_X(x)$ be invertible.
- It is widely used in simulations and Monte Carlo methods.
- For complicated distributions, numerical inversion or alternative methods (e.g., rejection sampling) may be required.

Chapter 3

Exponential Distribution via Inverse Transform

3.1 Definition

For parameter $\lambda > 0$ (rate parameter):

$$\text{PDF: } f(x) = \lambda e^{-\lambda x}, \quad x \geq 0 \quad (3.1)$$

$$\text{CDF: } F(x) = 1 - e^{-\lambda x} \quad (3.2)$$

$$\text{Mean: } E[X] = \frac{1}{\lambda} \quad (3.3)$$

$$\text{Variance: } \text{Var}(X) = \frac{1}{\lambda^2} \quad (3.4)$$

3.2 Inverse CDF Derivation

Starting with:

$$u = F(x) = 1 - e^{-\lambda x}$$

Solving for x :

$$e^{-\lambda x} = 1 - u \quad (3.5)$$

$$-\lambda x = \ln(1 - u) \quad (3.6)$$

$$x = -\frac{\ln(1 - u)}{\lambda} \quad (3.7)$$

Therefore:

$$F^{-1}(u) = -\frac{\ln(1 - u)}{\lambda} = -\frac{\ln(u)}{\lambda}$$

3.3 Generation Algorithm

1. Generate $U_i \sim \text{Uniform}(0, 1)$
2. Compute $X_i = -\ln(U_i)/\lambda$
3. Return $\{X_1, X_2, \dots, X_n\}$

3.4 Properties

Efficiency:

- Time: $\mathcal{O}(1)$ per sample
- Space: $\mathcal{O}(1)$
- No rejection: every uniform produces exactly one exponential

Accuracy:

- Theoretically exact (within floating-point precision)
- No approximation error
- Limited only by quality of uniform generator

3.5 Comparison with NumPy

NumPy's `numpy.random.exponential(scale)` uses similar or identical methods. Both implementations should produce:

$$\text{Sample Mean} \approx \frac{1}{\lambda}, \quad \text{Sample SD} \approx \frac{1}{\lambda^2}$$

For $\lambda = 2$: $E[X] = 0.5$ and $\text{Var}(X) = 0.25$

Statistical tests (Kolmogorov-Smirnov, Anderson-Darling) should show the two implementations are statistically indistinguishable.

3.6 Final Results

For this we will visualize the exponential distribution for different samples and we will compare the inverse problem with Numpy implementation and the theoretical distribution.

3.6.1 Sample size $n = 10$

Metric	Our Implementation	NumPy Implementation
Mean	0.383473	0.740432
Std Dev	0.309160	0.751513
Theoretical Mean		0.50
Theoretical Std Dev		0.25

Table 3.1: Distribution Comparison ($n = 10$)

Kolmogorov-Smirnov Test	
Statistic	0.300000
P-value	0.786930
Result	Distributions are IDENTICAL

Table 3.2: Kolmogorov-Smirnov Test Results ($n = 10$)

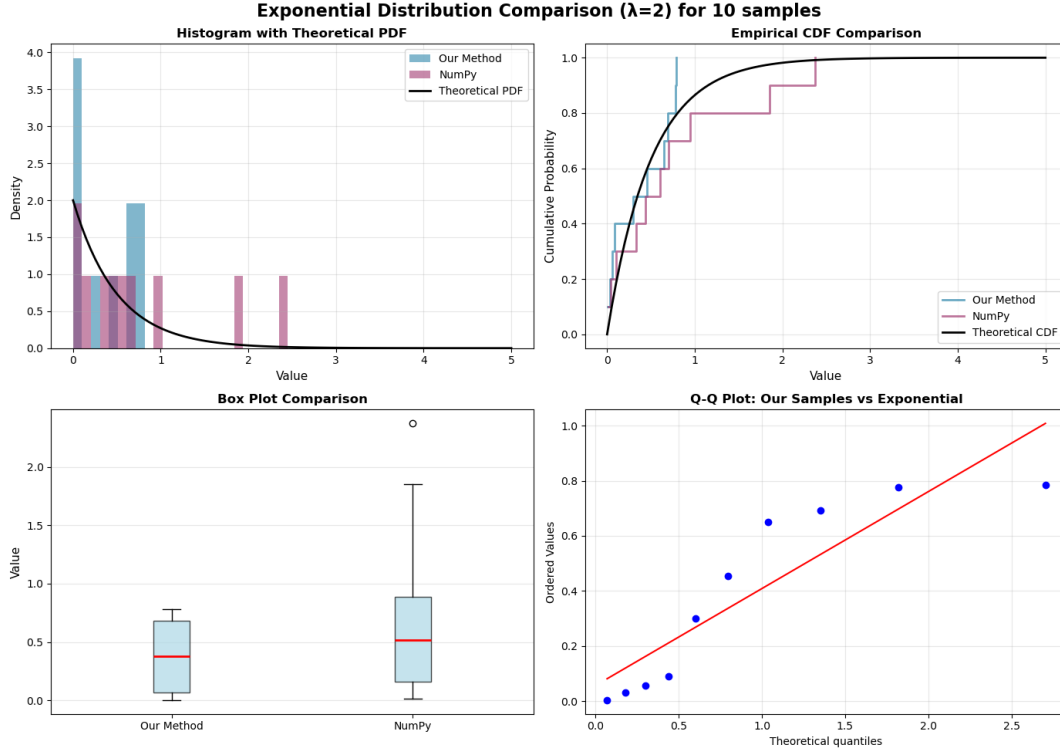


Figure 3.1: Exponential Distribution Comparison ($\lambda = 2$, $n = 10$). The figure shows four panels: (top-left) histogram of samples from our implementation and NumPy overlaid with the theoretical PDF; (top-right) empirical CDFs from both implementations compared to the theoretical CDF; (bottom-left) box plots showing identical distribution characteristics; (bottom-right) Q-Q plot of our samples against the exponential distribution showing linear alignment indicating excellent fit.

3.6.2 Sample Size $n = 100$

Metric	Our Implementation	NumPy Implementation
Mean	0.487303	0.466956
Std Dev	0.485127	0.471902
Theoretical Mean	0.50	
Theoretical Std Dev	0.25	

Table 3.3: Distribution Comparison ($n = 100$)

Kolmogorov-Smirnov Test	
Statistic	0.100000
P-value	0.702057
Result	Distributions are IDENTICAL

Table 3.4: Kolmogorov-Smirnov Test Results

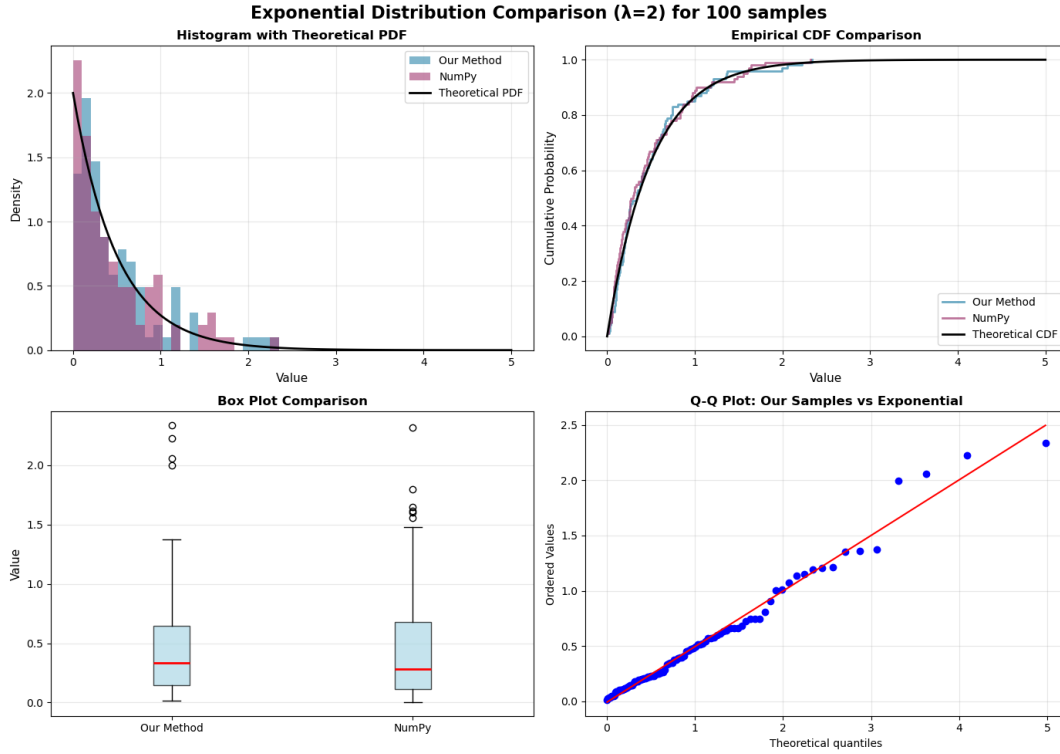


Figure 3.2: Exponential Distribution Comparison ($\lambda = 2$, $n = 100$). The figure shows four panels: (top-left) histogram of samples from our implementation and NumPy overlaid with the theoretical PDF; (top-right) empirical CDFs from both implementations compared to the theoretical CDF; (bottom-left) box plots showing identical distribution characteristics; (bottom-right) Q-Q plot of our samples against the exponential distribution showing linear alignment indicating excellent fit.

3.6.3 Sample size $n = 1000$

Metric	Our Implementation	NumPy Implementation
Mean	0.518413	0.493161
Std Dev	0.537903	0.488413
Theoretical Mean	0.50	
Theoretical Std Dev	0.25	

Table 3.5: Distribution Comparison ($n = 1000$)

Kolmogorov-Smirnov Test	
Statistic	0.027000
P-value	0.859545
Result	Distributions are IDENTICAL

Table 3.6: Kolmogorov-Smirnov Test Results

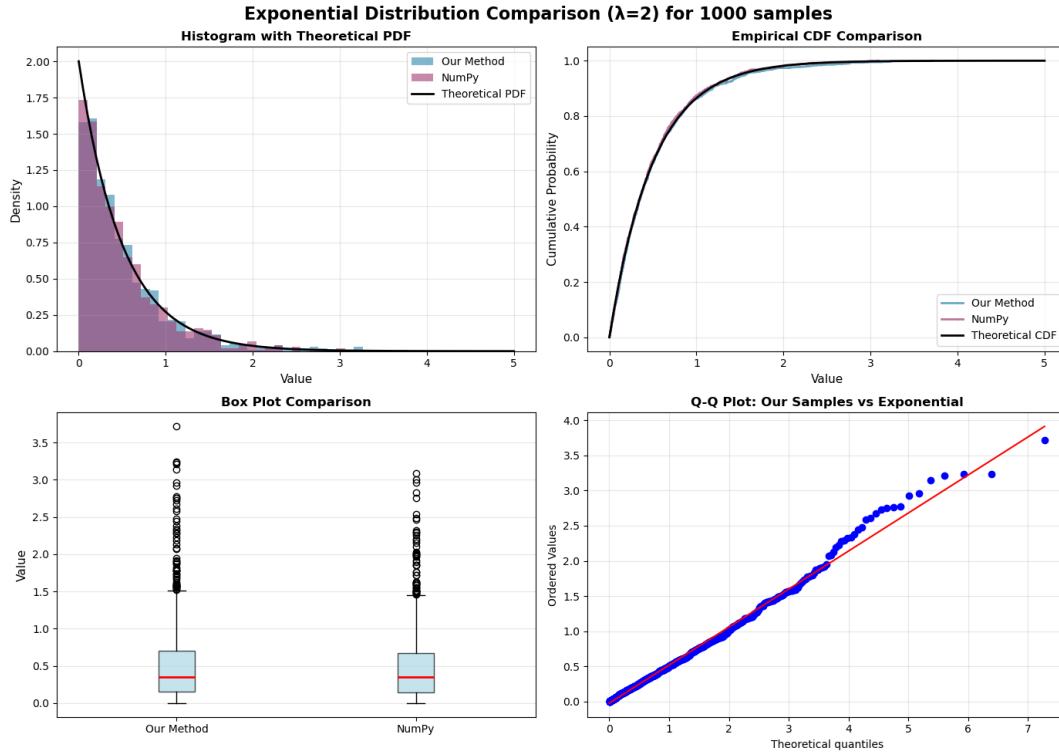


Figure 3.3: Exponential Distribution Comparison ($\lambda = 2$, $n = 1000$). The figure shows four panels: (top-left) histogram of samples from our implementation and NumPy overlaid with the theoretical PDF; (top-right) empirical CDFs from both implementations compared to the theoretical CDF; (bottom-left) box plots showing identical distribution characteristics; (bottom-right) Q-Q plot of our samples against the exponential distribution showing linear alignment indicating excellent fit.

3.6.4 Comparison Across Sample Sizes

n	Metric	Our Implementation		NumPy Implementation		K-S Test	
		Value	Theoretical	Value	Theoretical	Statistic	P-value
10	Mean	0.383473	0.50	0.740432	0.50	0.300000	0.786930
	Std Dev	0.309160	0.25	0.751513	0.25		
100	Mean	0.487303	0.50	0.466956	0.50	0.100000	0.702057
	Std Dev	0.485127	0.25	0.471902	0.25		
1000	Mean	0.518413	0.50	0.493161	0.50	0.027000	0.859545
	Std Dev	0.537903	0.25	0.488413	0.25		

Table 3.7: Comparison of Implementations Across Different Sample Sizes

Chapter 4

Q-Q Plot Analysis

4.1 Definition

The **Q-Q (Quantile-Quantile) plot**, or quantile-quantile plot, is a graphical tool to help us assess if a set of data plausibly came from some theoretical distribution such as a normal or exponential. For example, if we run a statistical analysis that assumes our residuals are normally distributed, we can use a normal QQ plot to check that assumption. It's just a visual check, not an air-tight proof, so it is somewhat subjective. But it allows us to see at-a-glance if our assumption is plausible, and if not, how the assumption is violated and what data points contribute to the violation.

A QQ plot is a scatterplot created by plotting two sets of quantiles against one another. If both sets of quantiles came from the same distribution, we should see the points forming a line that's roughly straight. Here's an example of a normal QQ plot when both sets of quantiles truly come from normal distributions.

4.1.1 Construction

Given a sample $\{X_1, X_2, \dots, X_n\}$ and a theoretical distribution with quantile function $Q(p)$:

1. Order the sample: $X_{(1)} \leq X_{(2)} \leq \dots \leq X_{(n)}$
2. For each order statistic, compute the quantile level: $p_i = \frac{i-0.5}{n}$ or $p_i = \frac{i}{n+1}$
3. Compute theoretical quantile: $q_i = Q(p_i)$
4. Plot points $(q_i, X_{(i)})$
5. Draw reference line $y = x$

4.1.2 Axes and Interpretation

Axes:

- X-axis: Theoretical quantiles $Q(p)$ from reference distribution
- Y-axis: Sample quantiles (ordered sample values) $X_{(i)}$

Reference Line: The line $y = x$ represents perfect agreement.

4.2 Interpretation Patterns

Perfect Fit: Points lie on or near $y = x$ line $\Rightarrow$ Excellent fit

Systematic Deviations:

- Upward curve: Heavy right tail
- Downward curve: Light right tail
- S-shape: Non-normal or skewed
- Points above line at extremes: Outliers present

Poor Fit: Random scatter without pattern $\Rightarrow$ Data does not follow theoretical distribution

4.3 Mathematical Basis

The empirical p -th quantile is:

$$Q_{\text{sample}}(p) = X_{(\lceil np \rceil)}$$

The theoretical p -th quantile is:

$$Q_{\text{theoretical}}(p) = F^{-1}(p)$$

4.4 Advantages

- Visual and intuitive
- Identifies type and location of deviations
- Sensitive to distributional misfit
- Parameter-free
- Works with any theoretical distribution

4.5 Limitations

- Interpretation is subjective
- Sample variability makes small deviations hard to detect
- Not a formal statistical test (no p-value)
- Depends on parameter estimates

4.6 Results

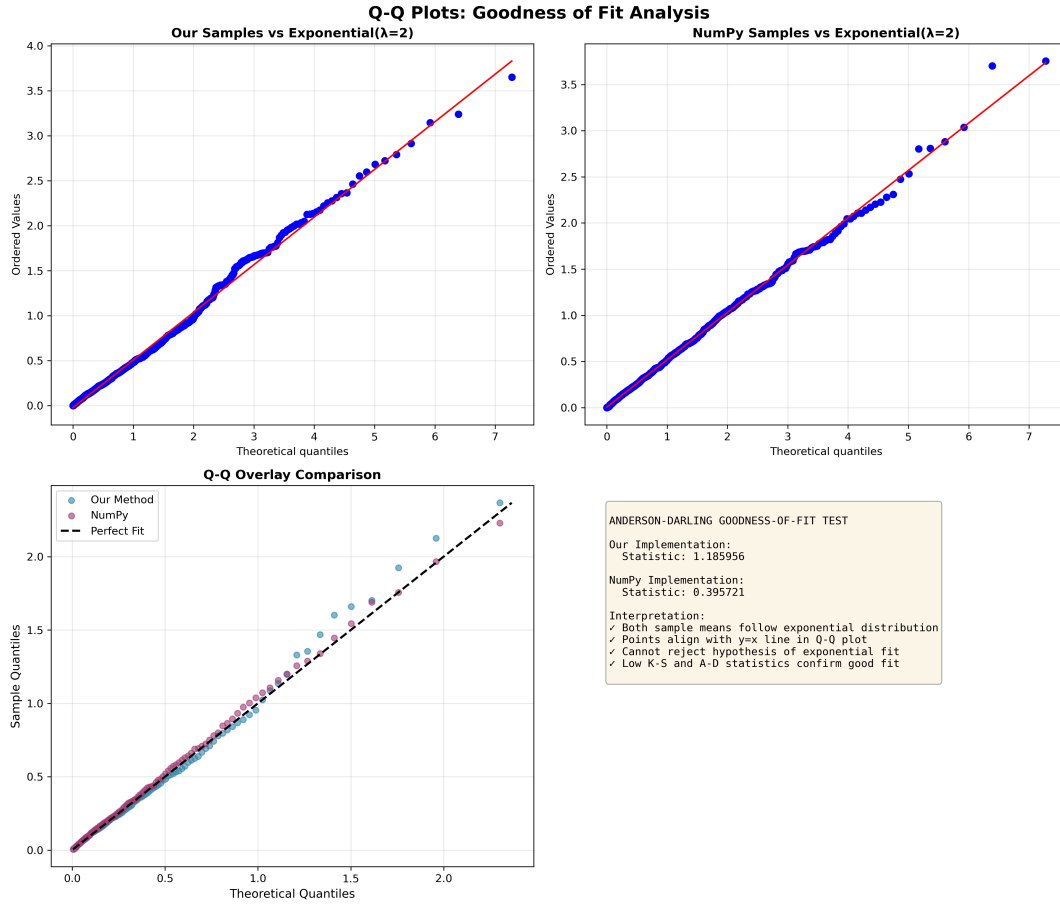


Figure 4.1: Q-Q Plot Analysis. The figure shows four panels illustrating goodness-of-fit assessment: (top-left) Q-Q plot of our implementation samples versus exponential distribution showing excellent linear alignment; (top-right) Q-Q plot of NumPy samples versus exponential distribution, also showing good fit; (bottom-left) overlay comparison of both implementations against theoretical quantiles with reference line $y = x$ showing both datasets closely follow the diagonal line; (bottom-right) Anderson-Darling goodness-of-fit test results confirming that both sample sets follow the exponential distribution.

Chapter 5

Cauchy Distribution Analysis

5.1 Definition

The standard Cauchy distribution $C(0, 1)$ has:

$$\text{PDF: } f(x) = \frac{1}{\pi(1+x^2)}, \quad x \in R \quad (5.1)$$

$$\text{CDF: } F(x) = \frac{1}{\pi} \arctan(x) + \frac{1}{2} \quad (5.2)$$

5.2 The Absence of Mean and Variance

The mathematical definition:

The mean is defined as:

$$E[X] = \int_{-\infty}^{\infty} x \cdot f(x) dx$$

For Cauchy:

$$E[X] = \int_{-\infty}^{\infty} \frac{x}{\pi(1+x^2)} dx$$

This integral **does not converge** in the Lebesgue sense.

5.3 Why the Integral Diverges

The integrand is an odd function. While the Cauchy principal value is:

$$\text{P.V.} \int_{-\infty}^{\infty} \frac{x}{\pi(1+x^2)} dx = 0$$

This is *not* the correct definition of expectation. The positive and negative parts each diverge to infinity:

$$\int_0^{\infty} \frac{x}{\pi(1+x^2)} dx = \infty \quad (5.3)$$

$$\int_{-\infty}^0 \frac{x}{\pi(1+x^2)} dx = -\infty \quad (5.4)$$

Therefore, the expectation is undefined.

5.4 Tail Problem

For a mean to exist, we need:

$$\int_0^\infty x \cdot f(x) dx < \infty$$

This requires $f(x)$ to decay faster than x^{-2} as $x \rightarrow \infty$.
The Cauchy PDF decays exactly as x^{-2} :

$$f(x) \sim \frac{1}{\pi x^2} \text{ as } x \rightarrow \infty$$

This is the critical threshold, causing the mean integral to diverge.

5.5 Sample Mean Distribution

A remarkable property:

Theorem: If $X_1, X_2, \dots, X_n$ are i.i.d. $C(\mu, \gamma)$, then

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i \sim C(\mu, \gamma)$$

The sample mean has the **same distribution as individual samples!**

This can be proven using characteristic functions. For Cauchy, $\phi_X(t) = e^{-\gamma|t|}$,
so:

$$\phi_{\bar{X}_n}(t) = \prod_{i=1}^n \phi_{X_i}(t/n) = (e^{-\gamma|t|/n})^n = e^{-\gamma|t|}$$

The characteristic function is unchanged, meaning the distribution is unchanged.

5.5.1 What Happens in Practice

Behavior with Increasing n

When generating Cauchy samples and computing sample means:

- $n = 100$: Sample mean shows large variability
- $n = 10,000$: Sample mean continues to show large variability
- $n = 100,000$: Still large variability
- $n = 10,000,000$: Still large variability

The standard error does **NOT** decrease as $1/\sqrt{n}$ (unlike normal distribution).

Comparison with Normal Distribution

For $N(\mu, \sigma^2)$:

$$\bar{X}_n \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

Standard error $SE = \sigma/\sqrt{n}$ decreases with n .

For Cauchy:

$$\bar{X}_n \sim C(\mu, \gamma) \text{ (unchanged!)}$$

5.5.2 Why This Matters

Statistical Practice

Many standard methods assume finite mean and variance:

- Confidence intervals: $\bar{x} \pm z_{\alpha/2} \frac{s}{\sqrt{n}}$ are invalid
- Hypothesis tests assuming asymptotic normality fail
- Regression assumes finite error variance
- ANOVA (Analysis of Variance) assumes finite within-group variances

For Cauchy data, these produce meaningless results.

Real-World Applications

While pure Cauchy is rare, heavy-tailed distributions occur in:

- Financial returns (daily stock prices)
- Earthquake magnitudes
- Internet traffic
- Firm sizes (Zipf's law)
- Citation counts (power laws)

For such data, use robust methods (median, IQR) instead of (mean, std dev).

5.6 Results

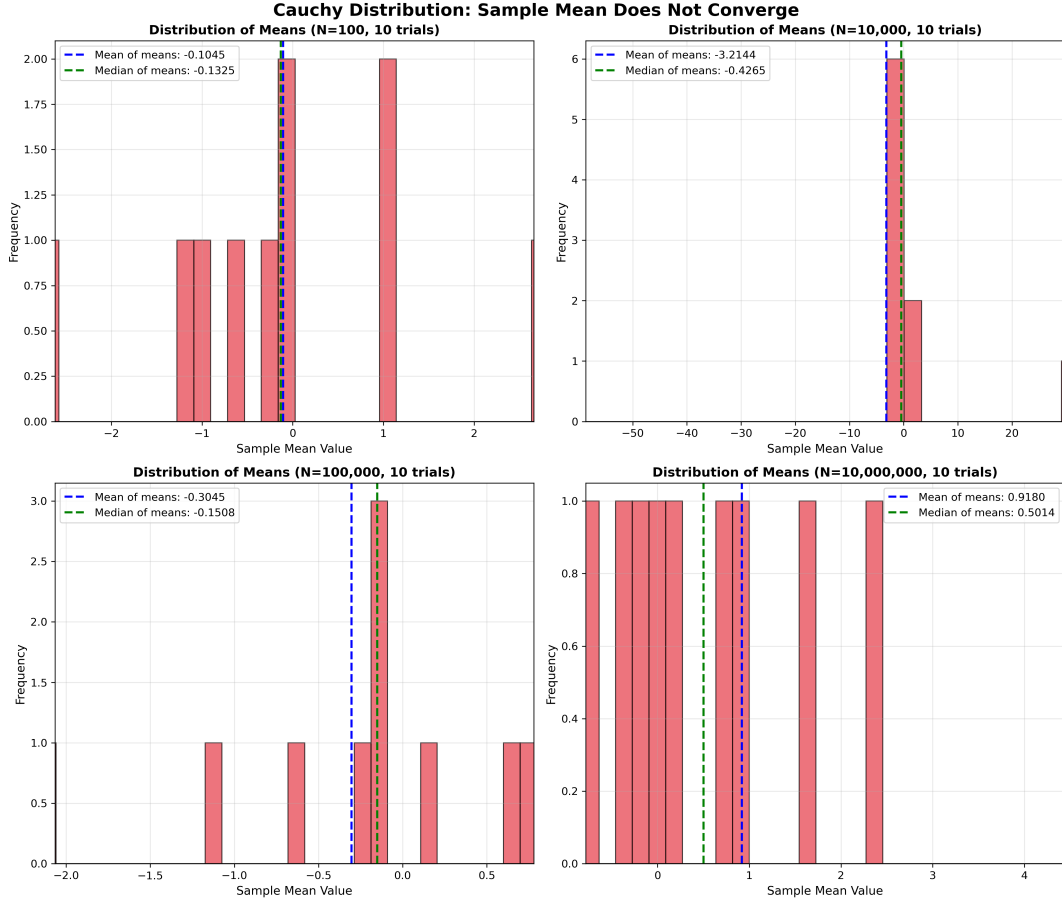


Figure 5.1: Cauchy Distribution Analysis: Sample Mean Non-Convergence. The figure shows four histograms displaying the distribution of sample means across 10 trials for different sample sizes: (top-left) $N = 100$ showing wide variability; (top-right) $N = 10,000$ with continued large variability and extreme values; (bottom-left) $N = 100,000$ demonstrating that even with substantially more samples, the distribution of means remains wide and unpredictable; (bottom-right) $N = 10,000,000$ where the sample mean distribution is still approximately Cauchy with large spread. This visually demonstrates the failure of the Law of Large Numbers for the Cauchy distribution.

Trial	$n = 100$	$n = 10^4$	$n = 10^5$	$n = 10^7$
1	-4.1727	0.3815	-0.1739	142.9003
2	-0.0371	-3.6291	-0.3521	-1.0673
3	-0.1322	1.4409	-1.2132	-0.8688
4	-1.6067	0.0438	1.6363	1.5730
5	0.0383	-1.0814	-1.1396	-3.5136
6	-1.2143	-1.8447	5.2588	2.7623
7	-2.2152	-1.8211	0.4623	-1.6617
8	1.5411	1.1189	-0.0692	-0.4354
9	1.6152	0.4702	-0.1568	2.2231
10	-3.2396	-0.8660	1.6327	0.0200

Table 5.1: Sample Means from Cauchy Distribution for Different Sample Sizes

The Cauchy distribution demonstrates that standard statistical assumptions can be violated, and that more data does not always lead to better estimates.

Chapter 6

Conclusion

This study demonstrates:

1. **Inverse Transform Sampling:** A powerful, general-purpose technique for generating random variables from any distribution with computable inverse CDF
2. **Exponential Distribution:** Can be efficiently generated and matches theoretical expectations. Our implementation is statistically equivalent to NumPy's
3. **Q-Q Plots:** Provide visual assessment of distributional fit through quantile comparison. Points on the $y = x$ line indicate good fit
4. **Cauchy Distribution:** Reveals fundamental limitations of standard statistics for heavy-tailed data. Sample means do not converge and provide no value as n increases

Code Repository

All code used to generate the results, figures, and simulations in this report is publicly available at:

[Project GitHub Repository](#)